Chapter 6:

Feature Selection

Feature Selection Techniques

- are the techniques to identify and select the most relevant and important features (independent variables) from a dataset.

- The primary objective of feature selection is to enhance the performance of a model by
  - mitigating overfitting,
  - improving generalization capabilities, and
  - reducing computational complexity.

Feature Selection Techniques

- Filter Methods
  - Correlation Coefficient
  - Chi-Square Test
  - Mutual Information
  - Variance Threshold

- Wrapper Methods
  - Recursive Feature Elimination (RFE)
  - Sequential Feature Selection (SFS

- Embedded Methods
  - Lasso Regression (L1 Regularization)
  - Ridge Regression (L2 Regularization)
  - Elastic Net

Feature Selection Techniques

Feature Selection Method	Description	Advantages	Disadvantages
Filter Methods	Select features based on statistical measures without involving a learning model.	- Fast and computationally efficient.	- May ignore feature interactions.
Wrapper Methods	Use a predictive model to evaluate subsets of features iteratively.	- Considers feature interactions. - Often yields better performance.	- Computationally expensive. - Prone to overfitting with small datasets.
Embedded Methods	Perform feature selection during the model training process.	- Integrated with the learning algorithm. - Efficient for large datasets.	- Limited to specific algorithms. - May not generalize well across models.

Method Aspect	Speed	Feature Interaction	Risk of Overfitting	Scalability to Big Data
Filter Methods	Fast	Not considered	Low	Excellent
Wrapper Methods	Slow	Considered	High	Poor
Embedded Methods	Medium/Fast (depends on model)	Considered	Medium	Good

Feature Selection vs Dimensionality Reduction

Feature Selection

- Feature selection focuses on identifying and retaining a subset of the most relevant or important features from the original dataset.
- The primary goal is to simplify the dataset by discarding irrelevant, redundant, or noisy features while preserving those that have the most predictive power or are highly correlated with the target variable

Dimensionality Reduction

- Dimensionality reduction transforms the original features into a new set of variables in a lower-dimensional space while retaining as much of the original information as possible.
- Generates new features that are combinations or projections of the original ones.

Feature Selection vs Dimensionality Reduction

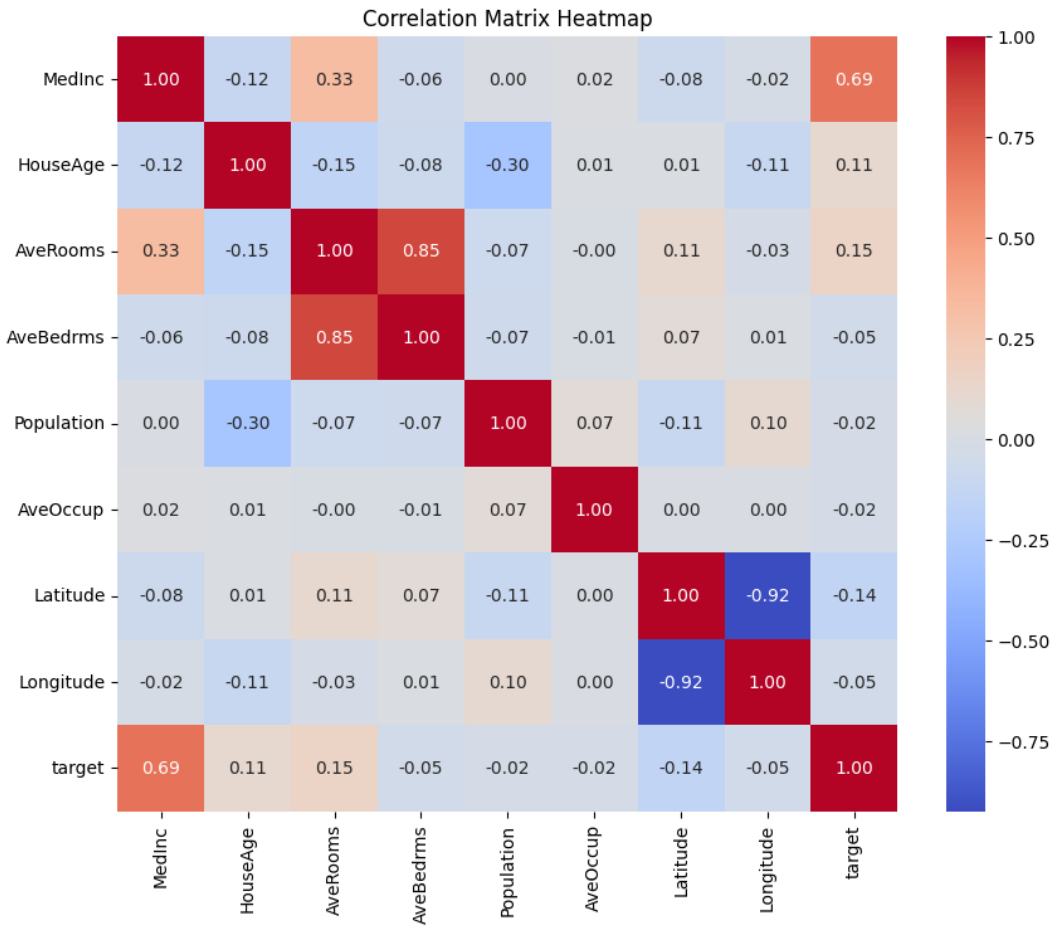
Aspect	Feature Selection	Dimensionality Reduction
Approach	Retains specific features from the original dataset.	Creates new features by transforming original ones.
Interpretability	Selected features are easily interpretable.	New dimensions may lack direct interpretability.
Purpose	Simplifies models by focusing on relevant variables.	Reduces dimensionality for visualization or efficiency.
Usage Context	Ideal when interpretability and simplicity are critical.	Useful for high-dimensional data requiring compact representations.

Filter Methods

Correlation Coefficient

Measures the linear relationship between features and the target variable. Features with high correlation to the target are selected.

- Identify correlations between independent variables and the dependent variable.



Selected Features with High Correlation to the Target:

MedInc 0.688075

Filter Methods

Feature	Correlation Coefficient (r)	Variance Inflation Factor (VIF)
Relationship	Between 2 variables (X1,X2)	Between 1 variable and a set (X1 vs. X2,X3,...Xn)
Objective	Measures strength/direction of association	Detects severity of multicollinearity
Interpretation	How do these two move together?	How "redundant" is this variable in the model?
Output	-1.0 to 1.0	1.0 to ∞

Filter Methods

Correlation Coefficient

Pearson correlation coefficient:

- This coefficient measures the linear relationship between two variables.
- It ranges from -1 to 1, where -1 indicates a perfect negative linear correlation, 1 indicates a perfect positive linear correlation, and 0 indicates no linear correlation.

Here's the formula for the Pearson correlation coefficient (r):

$$r = \frac{\sum(x_i - \hat{x})(y_i - \hat{y})}{\sqrt{\sum(x_i - \hat{x})^2 \sum(y_i - \hat{y})^2}}$$

where:

- x_i and y_i are the individual sample points
- x_i and y_i are the means of the x and y values, respectively

Filter Methods

$$r = \frac{\sum(x_i - \hat{x})(y_i - \hat{y})}{\sqrt{\sum(x_i - \hat{x})^2 \sum(y_i - \hat{y})^2}}$$

Pearson correlation coefficient calculation sample:

Let's say we have two variables, X and Y, with the following data points:

- $X = [1,2,3,4,5]$
- $Y = [2,4,5,4,5]$

1. Calculate the means:

$$\hat{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$\hat{y} = \frac{2 + 4 + 5 + 4 + 5}{5} = 4$$

Filter Methods

$$r = \frac{\sum(x_i - \hat{x})(y_i - \hat{y})}{\sqrt{\sum(x_i - \hat{x})^2 \sum(y_i - \hat{y})^2}}$$

Pearson correlation coefficient calculation sample:

- $X = [1,2,3,4,5]$
- $Y = [2,4,5,4,5]$

$\hat{x} = 3$

$\hat{y} = 4$

2. Calculate the numerator:

$$\sum(x_i - \hat{x})(y_i - \hat{y})$$

$= (1 - 3)(2 - 4) + (2 - 3)(4 - 4) + (3 - 3)(5 - 4) + (4 - 3)(4 - 4) + (5 - 3)(5 - 4)$

$= (-2)(-2) + (-1)(0) + (0)(1) + (1)(0) + (2)(1)$

$= 4+0+0+0+2$

$= \mathbf{6}$

Filter Methods

$$r = \frac{\sum(x_i - \hat{x})(y_i - \hat{y})}{\sqrt{\sum(x_i - \hat{x})^2 \sum(y_i - \hat{y})^2}}$$

Pearson correlation coefficient calculation sample:

- $X = [1,2,3,4,5]$
- $Y = [2,4,5,4,5]$

$\hat{x} = 3$

$\hat{y} = 4$

3. Calculate the denominator:

$$\sqrt{\sum(x_i - \hat{x})^2 \sum(y_i - \hat{y})^2}$$

$$\sum(x_i - \hat{x})^2 = (1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2$$

$$= 4 + 1 + 0 + 1 + 4$$

$$= \mathbf{10}$$

$$\sum(y_i - \hat{y})^2 = (2-4)^2 + (4-4)^2 + (5-4)^2 + (4-4)^2 + (5-4)^2$$

$$= 4 + 0 + 1 + 0 + 1$$

$$= \mathbf{6}$$

$$\text{Denominator} = \sqrt{10 \cdot 6} = \sqrt{60} \approx \mathbf{7.75}$$

Filter Methods

$$r = \frac{\sum(x_i - \hat{x})(y_i - \hat{y})}{\sqrt{\sum(x_i - \hat{x})^2 \sum(y_i - \hat{y})^2}}$$

Pearson correlation coefficient calculation sample:

4. Calculate the Pearson correlation coefficient:

$$r = \frac{6}{7.75} \approx \mathbf{7.77}$$

In this example, the Pearson correlation coefficient $r \approx 0.77$, which indicates a **strong positive** linear relationship between the variables X and Y.

Filter Methods

Chi-Square Test

used to determine if there is a significant association between two categorical variables.

- Calculate the observed and **expected frequencies** for each combination of feature values and target values.
- Compute the Chi-Square statistic using the formula:

$$x^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where

- O_i is the observed frequency.
- E_i is the expected frequency.

Filter Methods Chi-Square Test Calculation Example

$$x^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Assume we have the following dataset:

Feature	Target
A	1
A	0
B	1
B	1
B	0
C	1
C	0
C	0
C	0

- There are three categories (A, B, and C), so *i* takes values 1, 2, and 3 for these categories.
- The subscript *j* represents the column index in the contingency table

1. Contingency Table with *i* and *j* Indices:

Feature (*i*)	Target=1 (*j*=1)	Target=0 (*j*=2)	Row Total (*i*)
A (i = 1)	1	1	2
B (i = 2)	2	1	3
C (i = 3)	1	3	4
Column Total (j)	4	5	9

2. Expected Frequencies:

$$E_{ij} = \frac{(Row\ Total_i)(Column\ Total_j)}{Grand\ Total}$$

⇨

$$E_{11} = \frac{(2)(4)}{9} \approx 0.89 \qquad E_{12} = \frac{(2)(5)}{9} \approx 1.11$$

$$E_{21} = \frac{(3)(4)}{9} \approx 1.33 \qquad E_{22} = \frac{(3)(5)}{9} \approx 1.67$$

$$E_{31} = \frac{(4)(4)}{9} \approx 0.78 \qquad E_{32} = \frac{(4)(5)}{9} \approx 2.22$$

Filter Methods

$$x^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Chi-Square Test Example Calculation

Feature (*i*)	Target=1 (*j*=1)	Target=0 (*j*=2)	Row Total (*i*)
A (i = 1)	1	1	2
B (i = 2)	2	1	3
C (i = 3)	1	3	4
Column Total (j)	4	5	9

2. Expected Frequencies:

$E_{11} \approx 0.89$ $E_{12} \approx 1.11$
$E_{21} \approx 1.33$ $E_{22} \approx 1.67$
$E_{31} \approx 0.78$ $E_{32} \approx 2.22$

3. Chi-Square Statistic:

$$x^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where O_{ij} is the observed frequency and E_{ij} is the expected frequency.

Calculate for each cell:

$$x_{11}^2 = \frac{(1 - 0.89)^2}{0.89} \approx 0.014 \qquad x_{12}^2 = \frac{(1 - 1.11)^2}{1.11} \approx 0.011$$

$$x_{21}^2 = \frac{(2 - 1.33)^2}{1.33} \approx 0.336 \qquad x_{22}^2 = \frac{(1 - 1.67)^2}{1.67} \approx 0.268$$

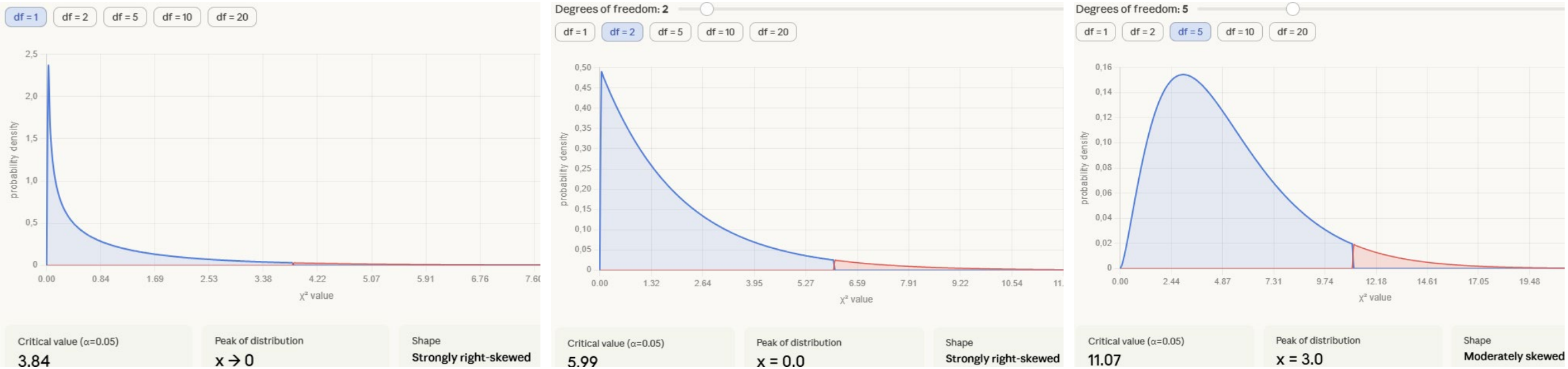
$$x_{31}^2 = \frac{(1 - 1.78)^2}{1.78} \approx 0.342 \qquad x_{32}^2 = \frac{(3 - 2.22)^2}{2.22} \approx 0.278$$

Sum all these values to get the total Chi-Square statistic:

$x^2 = 0.014 + 0.011 + 0.336 + 0.268 + 0.342 + 0.278 =$ **1.249**

Degrees of freedom = (number of rows − 1) × (number of columns – 1)

- Degrees of freedom refer to the number of independent values that are free to vary in a calculation, after certain constraints (like totals) are fixed.
- Tests whether observed data matches an expected distribution.



The red shaded area is the rejection region — where your χ^2 statistic must land to achieve $p < 0.05$. Notice how the critical value (the boundary of that red region) grows as df increases, meaning you need a larger χ^2

Significance Level

The significance level (denoted as α, alpha) is the threshold probability **you set <u>before</u>** running the test to decide when to reject the null hypothesis. It represents the maximum acceptable risk of making a Type I error — concluding there's an effect when there actually isn't one.

Alpha (α)	Meaning	Used In
0.05	5% risk of false positive	Most social/behavioral sciences
0.01	1% risk of false positive	Medical or stricter research
0.10	10% risk of false positive	Exploratory studies

The most commonly used value is $\alpha = 0.05$

Filter Methods

$$x^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Chi-Square Test Example Calculation

4. **Compare with Critical Value:**
   - Degrees of freedom: (number of rows − 1) × (number of columns – 1) = 2 × 1 = 2
   - Significance level (e.g., α = 0.05) critical value for 2 degrees of freedom is approximately 5.99.

Since our calculated Chi-Square statistic (1.249) is less than the critical value (5.99), we fail to reject the null hypothesis and conclude that there is no significant association between the feature and the target in this example.

Filter Methods

Chi-Square Test

Example if k = 2

```
data = {
    'Feature1': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Feature2': [10, 20, 10, 20, 10, 20, 10, 20, 10, 20],
    'Feature3': [1, 1, 2, 2, 3, 3, 4, 4, 5, 5],
    'Target': [0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
}
```

Output

```
Selected Feature Indices: [0 1]

Selected Feature Names: ['Feature1', 'Feature2']
```

Filter Methods

Mutual Information

quantifies the amount of information one random variable provides about another. Mutual Information filter is employed to evaluate the relevance and utility of individual features in relation to the target variable

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

where $p(x,y)$ is the joint probability distribution of X and Y, and $p(x)$and $p(y)$ are the marginal probability distributions of X and Y respectively.

Mutual Information Calculation Example

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

Consider a **binary classification dataset** where **Purchase Decision** depends on **Income Level**.

Person	Age Group	Income Level	Purchase Decision
A	Young	Low	No
B	Young	High	Yes
C	Middle	High	Yes
D	Senior	Low	No
E	Middle	Medium	Yes
F	Senior	High	Yes

Step 1: Compute Probability Values

Count the occurrences of each feature and target:
Marginal Probabilities

- P(Low) = 2 / 6 = 0.333
- P(Medium) = 1 / 6 = 0.167
- P(High) = 3 / 6 = 0.500
- P(No) = 2 / 6 = 0.333
- P(Yes) = 4 / 6 = 0.667

Joint Probabilities

- P(Low, No) = 2 / 6 = 0.333
- P(High, Yes) = 3 / 6 = 0.500
- P(Medium, Yes) = 1 / 6 = 0.167

Mutual Information Calculation Example

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

Step 2: Compute Mutual Information

Using the formula:

$$I(Income\ Level, Purchase\ Decision) = \sum p(x,y)\, log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

For each pair:

$$I(Low, No) = 0.333 \times \log\left(\frac{0.333}{0.333 \cdot 0.333}\right) = 0.333 \times \log(3) = 0.158$$

$$I(High, Yes) = 0.500 \times \log\left(\frac{0.500}{0.500 \cdot 0.667}\right) = 0.500 \times \log(1.5) = 0.176$$

$$I(Medium, Yes) = 0.167 \times \log\left(\frac{0.167}{0.167 \cdot 0.667}\right) = 0.167 \times \log(1.5) = 0.059$$

Summing these:
I = 0.158 + 0.176 + 0.059 = 0.393

Thus, **Income Level** contributes **0.393** bits of information about **Purchase Decision**.

Person	Age Group	Income Level	Purchase Decision
A	Young	Low	No
B	Young	High	Yes
C	Middle	High	Yes
D	Senior	Low	No
E	Middle	Medium	Yes
F	Senior	High	Yes

Marginal Probabilities

- P(Low) = 2 / 6 = 0.333
- P(Medium) = 1 / 6 = 0.167
- P(High) = 3 / 6 = 0.500
- P(No) = 2 / 6 = 0.333
- P(Yes) = 4 / 6 = 0.667

Joint Probabilities

- P(Low, No) = 2 / 6 = 0.333
- P(High, Yes) = 3 / 6 = 0.500
- P(Medium, Yes) = 1 / 6 = 0.167

Mutual Information Calculation Example

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

We now examine **Age Group** as a potential predictor of **Purchase Decision**.

Person	Age Group	Income Level	Purchase Decision
A	Young	Low	No
B	Young	High	Yes
C	Middle	High	Yes
D	Senior	Low	No
E	Middle	Medium	Yes
F	Senior	High	Yes

Step 1: Compute Probability Values

From the table, we estimate:
Marginal Probabilities

- P(Young) = 2 / 6 = 0.333
- P(Middle) = 2 / 6 = 0.333
- P(Senior) = 2 / 6 = 0.333
- P(No) = 2 / 6 = 0.333
- P(Yes) = 4 / 6 = 0.667

Joint Probabilities

- P(Young, No) = 1 / 6 = 0.167
- P(Young, Yes) = 1 / 6 = 0.167
- P(Middle, Yes) = 2/ 6 = 0.333
- P(Senior, No) = 1 / 6 = 0.167
- P(Senior, Yes) = 1 / 6 = 0.167

Mutual Information Calculation Example

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

Step 2: Compute Mutual Information

For each pair:

$$I(Young, No) = 0.167 \times \log\left(\frac{0.167}{0.333 \cdot 0.333}\right) = 0.167 \times \log(1.5) = 0.059$$

$$I(Young, Yes) = 0.167 \times \log\left(\frac{0.167}{0.333 \cdot 0.667}\right) = 0.167 \times \log(0.75) = -0.036$$

$$I(Middle, Yes) = 0.333 \times \log\left(\frac{0.333}{0.333 \cdot 0.667}\right) = 0.333 \times \log(1.5) = 0.118$$

$$I(Senior, No) = 0.167 \times \log\left(\frac{0.167}{0.333 \cdot 0.333}\right) = 0.167 \times \log(1.5) = 0.059$$

$$I(Senior, Yes) = 0.167 \times \log\left(\frac{0.167}{0.333 \cdot 0.667}\right) = 0.167 \times \log(0.75) = -0.036$$

Summing:

I = 0.059 + (−0.036) + 0.118 + 0.059 + (−0.036) = 0.164

Mutual Information Calculation Example

Result:

- **Age Group I = 0.164**, which is **lower** than **Income Level I = 0.393**.
- This suggests **Income Level is a better predictor** of Purchase Decision than Age Group.

Filter Methods

Variance Threshold

- designed to eliminate features that exhibit **low variance**, as these are unlikely to contribute substantially to the predictive capabilities of classification or regression models.

- the premise is that features with minimal variability offer limited information, thereby diminishing their utility in enhancing model performance.

Filter Methods

Variance Threshold Calculation Example

Consider a dataset with the following features and samples:

Sample	X_1	X_2	X_3
1	1	2	3
2	1	2	4
3	1	2	3
4	1	2	4
5	1	2	3

1. **Calculate the Variance for Each Feature:**
   - For X_1:

$$Variance(X_1) = \frac{1}{N}\sum_{i=1}^{N}(X_{1i} - \overline{X_1})^2$$

Since all values in X_1 are the same (1), the variance is 0.

- For X_2:

$$Variance(X_2) = \frac{1}{N}\sum_{i=1}^{N}(X_{2i} - \overline{X_2})^2$$

Since all values in X_2 are the same (2), the variance is 0.

- For X_3:

$$Variance(X_3) = \frac{1}{N}\sum_{i=1}^{N}(X_{3i} - \overline{X_3})^2$$

Calculate the mean of X_3:

$$\overline{X_3} = \frac{3+4+3+4+3}{5} = 3.4$$

Calculate the variance for $Variance(X_3)$

$$= \frac{1}{5}[(3 - 3.4)^2 + (4 - 3.4)^2 + (3 - 3.4)^2 + (4 - 3.4)^2 + (3 - 3.4)^2]$$

$$= \frac{1}{5}[0.16 + 0.36 + 0.16 + .36 + 0.16]$$

$$= \frac{1}{5} \times 1.2 = \mathbf{0.24}$$

Filter Methods

Variance Threshold Calculation Example

2. Set a Threshold and Select Features:

- Let's set a variance threshold of 0.1.
- Since Variance(X_1) = 0 and Variance(X_2) = 0 are both below the threshold, X_1 and X_2 will be removed.
- X_3 has a variance of 0.24, which is above the threshold, so X_3 will be retained.

Result: After applying the Variance Threshold method with a threshold of 0.1, we retain only the feature X_3.

Filter Methods

Method	Description	Type of Data	Advantages	Disadvantages	Use Cases
Correlation Coefficient	Measures the strength and direction of a linear relationship between two variables.	Continuous	- Simple and fast. - Useful for identifying linear relationships.	- Only captures linear dependencies. - Cannot handle categorical data.	- Feature selection for regression or continuous data analysis.
Chi-Square Test	Tests the independence between categorical features and the target variable.	Categorical	- Easy to use for categorical data. - Helps identify dependencies.	- Not applicable to continuous features. - Assumes independence of observations.	- Text classification - Categorical data analysis.
Mutual Information	Measures the amount of shared information between a feature and the target variable.	Categorical or Continuous	- Captures non-linear relationships. - Flexible for different data types.	- Computationally expensive for large datasets. - May require discretization.	- Any feature selection task, especially with non-linear relationships.
Variance Threshold	Selects features based on their variance (removes low-variance features that are unlikely to be informative).	Continuous or Categorical (if encoded)	- Simple and efficient. - Useful for removing redundant or uninformative features.	- Ignores the relationship with the target variable. - Threshold needs careful tuning.	- Preprocessing step in high-dimensional datasets.

Wrapper Methods

Recursive Feature Elimination (RFE)

is a technique that iteratively removes the least important features from a dataset and constructs a model using the remaining features.

This method ranks features based on their relative importance, typically assessed through metrics such as coefficients or permutation importance, and selects the optimal subset of features that maximizes model performance

Recursive Feature Elimination (RFE)

Calculation Example:

Scenario: Predicting Final Exam Scores

We have the following features and target variable:

Student ID	Study Hours (X_1)	Class Participation (X_2)	Assignment Scores (X_3)	Extracurricular Activities (X_4)	Final Exam Score (Y)
1	5	8	78	3	85
2	6	7	80	2	88
3	8	9	90	1	92
4	4	6	72	4	82
5	7	8	85	3	90

The goal is to predict **Final Exam Score (Y)** and select the **top 2 most important features** using RFE.

Recursive Feature Elimination (RFE) Calculation Example:

Step 1: Initial Model with All Features

First, train a regression model using all features (X_1, X_2, X_3, X_4). Assume the model provides the following coefficients as feature importance:

- X_1 (Study Hours): 3.5
- X_2 (Class Participation): 2.1
- X_3 (Assignment Scores): 1.8
- X_4 (Extracurricular Activities): 0.5

Step 2: Eliminate the Least Important Feature

The least important feature is X_4 (Extracurricular Activities) since its coefficient is the smallest (0.5). Remove X_4 from the feature set.

Step 3: Retrain the Model

Now, train the model using the remaining features (X_1, X_2, X_3). Assume the updated coefficients are:

- X_1: 4.0
- X_2: 2.5
- X_3: 2.0

Recursive Feature Elimination (RFE) Calculation Example:

Step 4: Eliminate Another Feature

The least important feature now is X_3 (Assignment Scores) with a coefficient of 2.0. Remove X_3 from the feature set.

Step 5: Final Model

Train the model using the remaining features (X_1, X_2):

- X_1: 4.2
- X_2: 2.8

Final Result

The top 2 features selected through RFE are:

1. **Study Hours (X_1)**
2. **Class Participation (X_2)**

Wrapper Methods

Sequential Feature Selection (SFS):

There are two primary variants of Sequential Feature Selection:

- Forward Selection: Begins with an empty set of features and iteratively adds the most informative feature to the model until a stopping criterion is met.

- Backward Elimination: Elimination starts with the full set of features and iteratively removes the least important feature until a predetermined number of features remains or a performance threshold is reached

Sequential Feature Selection (SFS) Calculation Example:

Scenario: Predicting Student Exam Scores

We have the following data:

Student ID	Study Hours (X_1)	Class Participation (X_2)	Assignment Scores (X_3)	Extracurricular Activities (X_4)	Exam Score (Y)
1	5	8	78	3	85
2	6	7	80	2	88
3	8	9	90	1	92
4	4	6	72	4	82
5	7	8	85	3	90

The goal is to predict **Exam Score (Y)** and identify the best subset of features using Sequential Feature Selection.

Sequential Feature Selection (SFS) Calculation Example:

Step 1: Train Model with No Features

Start with no features selected and calculate the model performance for each individual feature (X_1, X_2, X_3, X_4) separately.

Assume R^2 (performance score) for models with individual features:

- Model with X_1 (Study Hours): $R^2 = 0.85$
- Model with X_2 (Class Participation): $R^2 = 0.72$
- Model with X_3 (Assignment Scores): $R^2 = 0.80$
- Model with X_4 (Extracurricular Activities): $R^2 = 0.50$

Select the best-performing feature:

The best feature is X_1 (Study Hours), as it has the highest $R^2 = 0.85$.

Step 2: Add Second Feature

Now train models by combining X_1 with one additional feature (X_2, X_3, X_4).

Assume R^2 for combined models:

- **Model with $X_1 + X_2$:** $R^2 = 0.88$
- **Model with $X_1 + X_3$:** $R^2 = 0.92$
- **Model with $X_1 + X_4$:** $R^2 = 0.87$

Select the best-performing pair:

The best pair is $X_1 + X_3$, as it has the highest $R^2 = 0.92$.

Sequential Feature Selection (SFS) Calculation Example:

Step 3: Add Third Feature

Now train models by combining $X_1 + X_3$ with one additional feature (X_2, X_4).

Assume R^2 for combined models:

- **Model with $X_1 + X_3 + X_2$:** $R^2 = 0.94$
- **Model with $X_1 + X_3 + X_4$:** $R^2 = 0.91$

Select the best-performing combination:

The best combination is $X_1 + X_3 + X_2$, as it has the highest $R^2 = 0.94$.

Step 4: Stop When Performance No Longer Improves

At this stage, adding the fourth feature (X_4) might not significantly improve the performance. Thus, the selected features are:

- **Study Hours (X_1)**
- **Assignment Scores (X_3)**
- **Class Participation (X_2)**

Final Model

The final predictive model uses the features X_1, X_3, X_2 to predict Exam Score (Y).

Comparison Table

Feature	Recursive Feature Elimination (RFE)	Backward Sequential Feature Selection (BSFS)
Selection Criteria	Uses model internals (coefficients, feature importance, or feature ranking from a custom function).	Uses model performance (accuracy, AUC, RMSE, etc.) typically evaluated with cross-validation.
Computational Cost	Faster. It removes features in batches (or based on ranking) without retraining the model for every single candidate removal.	Slower. At each step, it must train the model k times (where k is the number of remaining features) to evaluate the impact of removing each one individually.
Stability	Can be unstable if the model's feature importance scores are unstable (e.g., multicollinear features cause coefficient variance in linear models).	More robust for complex interactions because it directly measures the impact of removal on the objective metric.
Model Dependence	The selected subset is biased toward the estimator used. A linear SVM RFE will yield a different subset than a Random Forest RFE.	The selected subset is dependent on the model used for evaluation, but the selection logic is generic. You can swap models easily within the same BSFS framework.
Handling Redundancy	May arbitrarily pick one feature from a highly correlated group based on minor coefficient differences.	May keep or remove correlated features based strictly on whether their removal hurts the objective metric.

Comparison

Step-by-Step Example
Imagine you have a dataset with 5 features: [A, B, C, D, E].

Using Backward Sequential Feature Selection (BSFS):

1.Evaluate All: Train the model on all 5 features. Get baseline performance (e.g., Accuracy = 0.92).
2.Iteration 1: For each feature f in [A, B, C, D, E], remove it temporarily.

1. Remove A: Train on [B, C, D, E] → Accuracy = 0.91.
2. Remove B: Train on [A, C, D, E] → Accuracy = 0.90.
3. Remove C: Train on [A, B, D, E] → Accuracy = **0.92**.
4. Remove D: Train on [A, B, C, E] → Accuracy = 0.85.
5. Remove E: Train on [A, B, C, D] → Accuracy = 0.89.

3.Decision: Removing C did not hurt accuracy (0.92 vs 0.92), so **C is permanently removed**.
4.Repeat with the remaining 4 features until a stopping criterion is met.

Using Recursive Feature Elimination (RFE):

1.Train Estimator: Train a model (e.g., Logistic Regression) on all 5 features.
2.Extract Ranking: Look at the model's coefficients (or feature importances). Suppose the absolute coefficients are:
A: 1.2, B: 1.1, C: 0.01, D: 2.5, E: 0.5.

1.Prune: The model internally decides that **C has the smallest coefficient** (0.01). RFE prunes C immediately (or marks it for elimination if using a batch size > 1).
2.Recurse: The model is retrained on [A, B, D, E], new coefficients are generated, and the feature with the new smallest coefficient is removed.

Wrapper Methods

Aspect	Recursive Feature Elimination (RFE)	Sequential Feature Selection (SFS)
Methodology	Iteratively removes the least important features based on model performance.	Iteratively adds (forward selection) or removes (backward elimination) features to optimize model performance.
Evaluation Approach	Evaluates feature importance using a predictive model (e.g., weights in linear models or feature importance in tree-based models).	Evaluates subsets of features incrementally to identify the optimal combination for the predictive model.
Direction of Selection	Backward elimination (starts with all features and removes the least relevant).	Can use forward selection (starts with no features and adds) or backward elimination (starts with all features and removes).
Computational Efficiency	Can be slower for large datasets due to repeated model training for every feature elimination.	Slower than RFE if evaluating all possible subsets; forward selection is generally faster than backward elimination.
Feature Interactions	Considers interactions among features as they are assessed within the model during each iteration.	May fail to fully capture feature interactions, especially in forward selection when starting with a minimal subset.
Use of Predictive Model	Requires a specific predictive model to rank and eliminate features.	Also requires a predictive model but evaluates subsets of features rather than ranks them directly.
Suitability for Large Datasets	Less suited for very high-dimensional data due to iterative removal.	Computationally expensive for large datasets or when using backward elimination.
Strengths	- Accounts for feature importance in the context of the model. - Works well with embedded model-specific metrics (e.g., feature importance in Random Forest).	- Flexible: Can adapt forward or backward approach. - Suitable for finding optimal subsets of features.
Weaknesses	- Computationally intensive for many features. - Dependent on the model used for ranking.	- Computationally expensive, especially for backward elimination. - Forward selection might miss optimal subsets due to incremental selection.
Examples	- Feature selection for tree-based models or regression models where feature ranking is available.	- Finding the best subset of features for models where feature importance is hard to measure directly (e.g., k-NN, SVM).

Embedded Methods

- are sophisticated techniques that integrate the process of feature selection directly into the model training phase.
- These methods effectively combine the advantages of both filter and wrapper methods, resulting in a more efficient and effective feature selection process.

Embedded Methods

Lasso Regression (L1 Regularization)

Lasso regression, formally known as the Least Absolute Shrinkage and Selection Operator

Shrinks the coefficients of less important features to zero, effectively performing feature selection.

$$Minimize\ \frac{1}{2}\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda\sum_{j=1}^{p}|\beta_j|$$

Here, y_i represents the response variable, x_{ij} are the predictor variables, β_j are the coefficients, and λ is the regularization parameter.

Lasso Regression (L1 Regularization) Calculation Example

Scenario: Predicting House Prices

We have the following sample dataset:

Square Footage (X_1)	Number of Bedrooms (X_2)	Distance to City Center (X_3)	House Price (Y)
1000	2	10	200,000
1500	3	15	250,000
2000	3	20	300,000
2500	4	25	350,000
3000	4	30	400,000

The objective is to predict **House Price (Y)** using Lasso Regression.

We aim to shrink less important coefficients to zero.

Lasso Regression (L1 Regularization) Calculation Example

Step 1: Standard Linear Regression

In standard linear regression, we minimize the sum of squared errors:

$$Cost\ Function = \sum (Y_i - \hat{Y}_i)^2$$

Where:

- Y_i: Actual house price.
- $\hat{Y}_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$: Predicted house price.

Assume Coefficients from Standard Linear Regression:

- $\beta_0 = 50{,}000$
- $\beta_1 = 100$
- $\beta_2 = 25$
- $\beta_3 = 5$

Step 2: Add Lasso Penalty

The Lasso cost function includes an additional penalty term:

$$Cost\ Function\ with\ Lasso = \sum (Y_i - \hat{Y}_i)^2 + \lambda \sum |\beta_j|$$

Where

- λ: Regularization strength (controls the amount of shrinkage).
- $\sum |\beta_j|$: Sum of the absolute values of the coefficients.

For example, if λ = 50,000, the penalty term becomes:

$\lambda \sum |\beta_j| = 50{,}000 \times (|100| + |25| + |5|) = 50{,}000 \times 130$

$= 6{,}500{,}000$

Lasso Regression (L1 Regularization) Calculation Example

Step 3: Shrink Coefficients

The Lasso algorithm shrinks the coefficients by balancing the trade-off between minimizing the errors and the penalty term.

How β Coefficients are Adjusted

The Lasso algorithm balances minimizing the residuals and the penalty term. This is achieved through **soft thresholding**—a mathematical approach that shrinks coefficients towards zero or eliminates them entirely.

1. Gradient Descent for Optimization

Lasso uses an iterative optimization process, often **gradient descent**, to calculate new β values:

- Start with initial β values (e.g., from ordinary least squares regression).
- Adjust β values iteratively based on:
  - The residuals $(Y_i - \hat{Y}_i)^2$.
  - The L1 penalty $\lambda|\beta_j|$, which applies shrinkage to each β_j.

2. Soft Thresholding

For each feature X_j:

- If the ordinary least squares solution is smaller than λ, set $\beta_j = 0$ (eliminate the feature).
- If larger, shrink β_j by subtracting λ.

The update rule for Lasso is:

$$\beta_j^{new} = sign(\beta_j^{old}) \,.\, max(|\beta_j^{old}| - \lambda, 0)$$

Where:

- $sign(\beta_j^{old})$: Maintains the direction (positive or negative) of the coefficient.
- $max(|\beta_j^{old}| - \lambda, 0)$: Shrinks coefficients towards zero.

Let's calculate new β\beta values for our example:

Initial Coefficients (Ordinary Least Squares):

- $\beta_1 = 100$
- $\beta_2 = 25$
- $\beta_3 = 5$

Lasso Regression (L1 Regularization) Calculation Example

Regularization Strength (λ):

Let's assume λ = 10.

$$\beta_j^{new} = sign(\beta_j^{old}) \,.\, max(|\beta_j^{old}| - \lambda, 0)$$

- $\beta_1 = 100$
- $\beta_2 = 25$
- $\beta_3 = 5$

Apply Soft Thresholding:

1. **For β_1:**

$$\beta_1^{new} = \text{sign}(100) \,.\, \max(|100| - 10, 0) = 1 \,.\, 90 = 90$$

1. **For β_2:**

$$\beta_2^{new} = \text{sign}(25) \,.\, \max(|25| - 10, 0) = 1 \,.\, 15 = 15$$

1. **For β_3:**

$$\beta_3^{new} = \text{sign}(5) \,.\, \max(|5| - 10, 0) = 1 \,.\, 0 = 0$$

Final Coefficients:

- $\beta_1^{new} = 90$
- $\beta_2^{new} = 15$
- $\beta_3^{new} = 0$ (eliminated due to shrinkage).

Impact of Regularization

1. **Lower** λ: Smaller shrinkage, coefficients are closer to original values.
2. **Higher** λ: Larger shrinkage, coefficients may shrink to zero, effectively performing feature selection.

Lasso Regression (L1 Regularization) Calculation Example

Step 4: Final Lasso Regression Equation

The Lasso regression equation becomes:

$$\hat{Y} = 55{,}000 + 90X_1 + 15X_2$$

Notice that X_3 (Distance to City Center) is no longer part of the equation because its coefficient was shrunk to zero.

Step 5: Prediction Example

Using the Lasso equation, predict the house price for:

- Square Footage (X_1) = 1800
- Number of Bedrooms (X_2) = 3

$$\hat{Y} = 55{,}000 + 90(1800) + 15(3)$$
$$\hat{Y} = 55{,}000 + 162{,}000 + 45$$

The predicted house price is **217,045**.

Embedded Methods

Ridge Regression (L2 Regularization)

Ridge regression, also referred to as Tikhonov regularization

Shrinks coefficients but does not set them to zero. Often used in combination with other methods.

$$Minimize\ \frac{1}{2}\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda\sum_{j=1}^{p}{\beta_j}^2$$

Here, y_i represents the response variable, x_{ij} are the predictor variables, β_j are the coefficients, and λ is the regularization parameter.

Ridge Regression (L2 Regularization) Calculation Example

Scenario: Predicting House Prices

We have the following sample dataset:

Square Footage (X_1)	Number of Bedrooms (X_2)	Distance to City Center (X_3)	House Price (Y)
1000	2	10	200,000
1500	3	15	250,000
2000	3	20	300,000
2500	4	25	350,000
3000	4	30	400,000

The goal is to predict **House Price (Y)** using Ridge Regression and understand how coefficients are adjusted.

Ridge Regression (L2 Regularization) Calculation Example

Step 1: Standard Linear Regression

In ordinary least squares (OLS) regression, we minimize the residual sum of squares:

$$Cost\ Function\ (OLS) = \sum (Y_i - \hat{Y}_i)^2$$

Let's assume the OLS regression gives the following coefficients:

- $\beta 0 = 50{,}000$
- $\beta_1 = 100$
- $\beta_2 = 25$
- $\beta_3 = 5$

Step 2: Ridge Regression

Ridge Regression modifies the cost function to include an **L2 penalty** term:

$$Cost\ Function\ (Ridge) = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

Where:

- λ: Regularization strength (controls the amount of shrinkage).
- $\sum_{j=1}^{p} \beta_j^2$: Sum of squared coefficients (L2 norm).

Ridge Regression (L2 Regularization) Calculation Example

Step 3: Adjust Coefficients with Ridge Regularization

For each feature X_j, Ridge adjusts the coefficients by balancing two terms:

1. **Residual Errors ($\sum(Y_i - \hat{Y}_i)^2$):**
   - Ridge aims to minimize prediction errors, just like OLS.
2. **Penalty Term ($\sum_{j=1}^{p} \beta_j^2$):**
   - Ridge shrinks the coefficients towards zero but does not eliminate them entirely (unlike Lasso).

The new coefficients are calculated by solving:

OLS Solution:

Let's assume OLS gives the following coefficients:

$$\beta^{OLS} = \begin{bmatrix} 100 \\ 25 \\ 5 \end{bmatrix}$$

Ridge Adjustment:

If λ = 10, adjust each coefficient:

$$\beta_j^{Ridge} = \frac{\beta_j^{OLS}}{1 + \lambda}$$

1. **For β_1 (Square Footage):**

$$\beta_1^{Ridge} = \frac{100}{1 + 10} = \frac{100}{11} \approx 9.09$$

2. **For β_2 (Number of Bedrooms):**

$$\beta_2^{Ridge} = \frac{25}{1 + 10} = \frac{25}{11} \approx 2.27$$

3. **For β_3 (Distance to City Center):**

$$\beta_3^{Ridge} = \frac{5}{1 + 10} = \frac{5}{11} \approx 0.45$$

Ridge Regression (L2 Regularization) Calculation Example

Final Coefficients:

- $\beta_1 = 9.09$
- $\beta_2 = 2.27$
- $\beta_3 = 0.45$

Step 4: Ridge Regression Equation

The updated Ridge regression equation becomes:

$$\hat{Y} = 50{,}000 + 9.09X_1 + 2{,}27X_2 + 0{,}45X_3$$

Step 5: Prediction Example

Let's predict the house price for:

- Square Footage (X_1) = 1800
- Number of Bedrooms (X_2) = 3
- Distance to City Center (X_3) = 20

Substitute the values into the Ridge equation:

$$\hat{Y} = 50{,}000 + 9.09(1800) + 2{,}27(3) + 0{,}45(20)$$

$$\hat{Y} = 50{,}000 + 16{,}362 + 6.81 + 9$$

The predicted house price is **66,377.81**.

Embedded Methods

Elastic Net:

combines the properties of both Lasso (Least Absolute Shrinkage and Selection Operator) and Ridge regression, thereby addressing some of the limitations inherent in Lasso while retaining its advantages.

This method integrates both L1 (Lasso) and L2 (Ridge) penalties into the linear regression objective function, .

$$Minimize\ \frac{1}{2}\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda_1 \sum_{j=1}^{p}|\beta_j| + \lambda_2 \sum_{j=1}^{p}{\beta_j}^2$$

Here, y_i represents the response variable, x_{ij} are the predictor variables, β_j are the coefficients, and λ_1 and λ_2 are regularization parameters controlling the L1 and L2 penalties, respectively.

Lasso Regression (L1 Regularization) Calculation Example

Scenario: Predicting House Prices

We have the following sample dataset:

Square Footage (X_1)	Number of Bedrooms (X_2)	Distance to City Center (X_3)	House Price (Y)
1000	2	10	200,000
1500	3	15	250,000
2000	3	20	300,000
2500	4	25	350,000
3000	4	30	400,000

We'll apply **Elastic Net Regularization** to identify the most important features for predicting **House Price (Y)**.

Lasso Regression (L1 Regularization) Calculation Example

Step 3: Apply Elastic Net Regularization

Parameters:

Let's assume: (found by hyperparameter tunning)

- λ = 10 (regularization strength).
- α = 0.5 (equal mix of L1 and L2 penalties).

Compute the Adjusted Coefficients:

Elastic Net modifies each coefficient β_j as follows:

1. **Apply L2 Regularization (Ridge Term)**: Shrink each coefficient:

$$\beta_j^{L2} = \frac{\beta_j^{OLS}}{1 + \lambda(1 - \alpha)}$$

2. **Apply L1 Regularization (Lasso Term)**: Use **soft thresholding**:

$$\beta_j^{Elastic\ Net} = sign(\beta_j^{L2}) \cdot max(|\beta_j^{L2}| - \lambda\alpha, 0)$$

Lasso Regression (L1 Regularization) Calculation Example

Step 4: Calculate Adjusted Coefficients

For $\boldsymbol{\beta_1}$ (Square Footage):

1. L2 shrinkage:

$$\beta_1^{L2} = \frac{120}{1+10(1-0.5)} = \frac{120}{1+5} = \frac{120}{6} = 20$$

2. L1 penalty:

$$\beta_1^{Elastic\ Net} = sign(20) \cdot max(|20| - 10 \cdot 0.5, 0) = 1 \cdot (max(20-5,0) = 15$$

For $\boldsymbol{\beta_2}$ (Square Footage):

1. L2 shrinkage:

$$\beta_2^{L2} = \frac{50}{1+10(1-0.5)} = \frac{50}{1+5} = \frac{50}{6} \approx 8.33$$

2. L1 penalty:

$$\beta_2^{Elastic\ Net} = sign(8.33) \cdot max(|8.33| - 5, 0) = 1 \cdot (max(8.33-5,0) = 3.33$$

For $\boldsymbol{\beta_3}$ (Square Footage):

1. L2 shrinkage:

$$\beta_3^{L2} = \frac{15}{1+10(1-0.5)} = \frac{15}{1+5} = \frac{15}{6} = 2.5$$

2. L1 penalty:

$$\beta_3^{Elastic\ Net} = sign(2.5) \cdot max(|2.5| - 5, 0) = 1 \cdot (max(2.5-5,0) = 0$$

Lasso Regression (L1 Regularization) Calculation Example

Step 5: Final Coefficients

After Elastic Net regularization, the adjusted coefficients are:

- $\beta_1 = 15$ (Square Footage)
- $\beta_2 = 3.33$ (Number of Bedrooms)
- $\beta_3 = 0$ (Distance to City Center)

Step 6: Feature Selection

- **Selected Features:** Elastic Net sets $\beta_3 = 0$, eliminating **Distance to City Center** as a feature since it does not significantly contribute to predicting house prices.
- **Retained Features:**
  - X_1: Square Footage ($\beta_1 = 15$).
  - X_2: Number of Bedrooms ($\beta_2 = 3.33$).

Embedded Methods

Aspect	Lasso Regression (L1 Regularization)	Ridge Regression (L2 Regularization)	Elastic Net
Methodology	Penalizes the absolute magnitude of coefficients (L1 norm), forcing some coefficients to be exactly zero (feature selection).	Penalizes the squared magnitude of coefficients (L2 norm), shrinking coefficients but retaining all features.	Combines L1 and L2 regularization penalties, balancing feature selection and coefficient shrinkage.
Purpose	Feature selection by eliminating irrelevant features (coefficients shrink to zero).	Reduces multicollinearity and prevents overfitting while keeping all features.	Achieves both feature selection and robust regularization.
Sparse Models	Produces sparse models as irrelevant features are excluded (coefficients = 0).	Retains all features with non-zero coefficients, even if their contributions are small.	Produces sparse models (like Lasso) while preventing over-aggressive shrinking.
Feature Selection	Yes (irrelevant features are excluded).	No (features are shrunk but not eliminated).	Yes (with controlled feature elimination and shrinkage).
Strengths	- Selects relevant features effectively. - Handles high-dimensional data well. - Interpretable due to sparse coefficients.	- Helps with multicollinearity. - Works well when all features are important. - Stable coefficients.	- Combines advantages of Lasso and Ridge. - Flexible in balancing selection and regularization.
Weaknesses	- May perform poorly with correlated features. - Aggressive regularization can over-simplify the model.	- Does not perform feature selection. - Over-shrinks coefficients in highly correlated data.	- Requires tuning of two hyperparameters (alpha and lambda). - Complex compared to Lasso or Ridge.
Use Cases	- Feature selection in high-dimensional datasets. - Sparse regression models (e.g., text classification).	- Regression problems with multicollinearity (e.g., financial or biological data).	- Data with correlated features requiring both selection and shrinkage. - Flexible modeling scenarios.